

Abstract algebra – 6

The hardest problems

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Theoretical minimum. Everything needed has already been proved in Abstract algebra – 1–5; what follows is the list of moves to try when a problem does not yield to the first idea.

Fact 1 (count the same thing twice). The class equation, the orbit–stabiliser theorem and the pairing $x \leftrightarrow x^{-1}$ are the three standard double counts. A hard group problem is usually a statement about a number that is computed in two ways, one of which is obviously divisible by p .

Fact 2 (get down to a field). A finite ring without zero divisors is a field (Theorem 1 and Wedderburn in Abstract algebra – 2); a finite commutative ring is a product of local rings; and any ring maps onto A/M with M maximal. Reducing modulo a maximal ideal turns an identity in A into an identity in a field, where linear algebra is available.

Fact 3 (linearise the group). Replacing G by the space of functions on G converts combinatorics into linear algebra: convolution, the averaging operator of Abstract algebra – 5, and, for abelian G , the Fourier transform over the character group (the discrete case is worked out in Fourier analysis – 4). Statements about sumsets, about the number of representations $g = xy$, and about random walks on G all become statements about eigenvalues.

Fact 4 (what may be quoted). Lagrange, Cauchy, the class equation, the classification of finite abelian groups, Wedderburn’s little theorem, Jacobson’s commutativity theorem, Maschke, Schur, the simplicity of A_n for $n \geq 5$, and Sylow’s theorems. Anything else should be proved.

Fact 5 (the three traps). Non-commutativity: a one-sided inverse need not be two-sided unless the ring is finite. Characteristic two: $x = -x$, and half of the identities collapse. Infinity: pigeonhole, averaging and “injective

implies surjective” all fail at once, and a problem about infinite groups is usually about that failure.

1. Let $c \geq 1$ be real, let G be an abelian group and let $A \subseteq G$ be finite with $|A+A| \leq c|A|$, where $X+Y = \{x+y\}$. Prove that $|\underbrace{A + A + \cdots + A}_k| \leq c^k|A|$ for every positive integer k . (IMC 2013)
2. Prove that a non-trivial finite ring is *not* a division ring if and only if the equation $x^n + y^n = z^n$ has a non-trivial solution in it for every natural number n . (Румыния, 1999)
3. Let n be a positive integer and $\mathbb{Z}_n$ the ring of integers modulo n . Suppose there is a function $f: \mathbb{Z}_n \rightarrow \mathbb{Z}_n$ with (i) $f(x) \neq x$, (ii) $f(f(x)) = x$, (iii) $f(f(f(x+1)+1)+1) = x$ for all x . Prove that $n \equiv 2 \pmod{4}$. (IMC)
4. Say that an abelian group G has property (P) if for every commutative group H , every subgroup $H' \leq H$ and every monomorphism $\mu': H' \rightarrow G$ there is a morphism $\mu: H \rightarrow G$ extending μ' . Show that $(\mathbb{Q}^*, \cdot)$ does not have property (P) and that $(\mathbb{Q}, +)$ does. (Румыния, 2000)
5. Let g, h be two distinct elements of a group G and let $n \geq 1$. Let $w = (w_1, w_2, \dots)$ be a sequence, not eventually periodic, with each $w_i \in \{g, h\}$, and let H be the subgroup generated by all products $w_k w_{k+1} \cdots w_{k+n-1}$, $k \geq 1$. Prove that H does not depend on the choice of w (though it may depend on n). (IMC)